

Giorgio De Santis

Robotics & AI Engineer | Perception, Control & Computer Vision
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Professional Summary

M.Sc. student in Artificial Intelligence and Robotics at Sapienza University of Rome (expected March 2027), with a B.Sc. in Electronic Engineering and a thesis on ML applications in electromagnetics. Comfortable across the robotics-AI stack: deriving and tuning impedance/force controllers for industrial manipulators, building ROS 2 navigation and social-interaction stacks, and training transformer-based vision models on satellite imagery, all resting on an electronics foundation of microcontrollers, DSP, and circuit design. Outside coursework, building and shipping small production systems end to end – including a fully automated content pipeline running on a daily cron schedule – and exploring computational-physics side projects (Ising models, LHC event classification) purely out of curiosity.

Education

M.Sc., Artificial Intelligence and Robotics — Sapienza University of Rome

Expected graduation: March 2027

Relevant coursework: Neural Networks, Robotics 1 & 2, Control Theory, Computer Vision, Model Predictive Control

B.Sc., Electronic Engineering — Sapienza University of Rome (2024)

Thesis: “Machine Learning Applications in Electromagnetics” | Focus: microcontroller architecture, digital signal processing, embedded C/C++, circuit theory, automatic controls – the hardware-and-signals foundation that now underpins my approach to embedded robotics and control (see Robotics & Control projects below)

Technical Skills

Languages: Python, C++, C, MATLAB, Lua

Robotics: ROS 2, Gazebo, RViz, CoppeliaSim, impedance/force control, null-space projection, obstacle avoidance, (LPV-)MPC

AI / ML: PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, Hugging Face, TorchGeo, Swin Transformer, U-Net, CNN, LSTM, random forests

Embedded & Electronics: STM32, Arduino, LTspice (+ TINA-TI)

Platforms & Tools: Git, Linux, Docker, LaTeX

Robotics & Control Projects

Dynamic Impedance Control for Teleoperated Ultrasound Probe — Python, CoppeliaSim

- Designed and tuned an impedance controller in Python for a 7-DOF KUKA LBR iiwa arm to replace rigid position control with compliant, force-limited teleoperation.
- Built and validated the full teleoperation and control loop directly in CoppeliaSim, achieving stable, safe probe-surface contact throughout simulated ultrasound-scanning trajectories.

Null-Space Obstacle Avoidance for Redundant Manipulators — MATLAB, CoppeliaSim, KUKA LBR iiwa 7

- Exploited the kinematic redundancy of the 7-DOF KUKA LBR iiwa to project obstacle-avoidance motion into the manipulator's null space while tracking a fixed Cartesian end-effector trajectory, keeping the task and the avoidance behavior decoupled.

LPV-MPC Control of an Underactuated Hexarotor — ROS 2, Gazebo

- Implementing a Linear-Parameter-Varying Model Predictive Control (LPV-MPC) scheme in ROS 2/Gazebo to decouple and stabilize the translational and rotational (Zero-Moment-Point) dynamics of an underactuated hexarotor UAV, as the capstone project for the Aerial Robotics course.

UAV Obstacle Avoidance via Vortex Vector Fields — Python, CoppeliaSim

- Contributed to a vortex-vector-field navigation scheme for UAV obstacle avoidance, generating smooth, collision-free trajectories around obstacles in cluttered simulated environments.

CareAssist-HRI: Cognitive Social Robot (early stage) — ROS 2

- Designing social costmaps and interaction logic so a socially-aware robot navigates and interacts appropriately around people, not just static obstacles – work in progress, not yet complete.

AI & Computer Vision Projects

Satellite Image Segmentation with Swin Transformer & U-Net — PyTorch, TorchGeo

- Implemented and benchmarked U-Net and Swin Transformer segmentation architectures for high-resolution semantic land-cover segmentation.
- Initialized backbones with SatMAE++ self-supervised pretrained weights and measured the effect of remote-sensing-specific pretraining against training from scratch, as the Neural Networks course project.

CNN vs. LSTM: Architecture Follows Data Shape — PyTorch

- Built a CNN to read the spatial texture of 2D Ising spin-model configurations and an LSTM to read the temporal order of six air-quality sensor streams, using the same regression framework to show how architecture choice should follow the shape of the data rather than the field it comes from.

Physics-Feature Trees vs. Raw-Image CNNs on LHC Events — scikit-learn, PyTorch

- Compared a random-forest classifier trained on hand-engineered physics features against a CNN trained on raw 32x32 detector images, on the same set of high-energy-physics collision events, to quantify what the raw-image model learns beyond expert features.

Dental Imagery: Synthetic Generation & Segmentation — PyTorch, computer vision

- Contributed to a deep-learning pipeline for synthetic generation and semantic segmentation of dental imagery, aimed at augmenting scarce labeled medical-imaging data.

Software Engineering & Personal Projects

Qultura – Automated Content Pipeline — Python, GitHub Actions, PWA

- Designed and built, end to end, a personal system that filters and organizes high-signal content on a daily schedule.
- Engineered an audio-transcription pipeline (yt-dlp + Groq Whisper API) running on an external cloud VM, feeding transcribed text into a digest and study-module builder.
- Automated the entire pipeline with scheduled GitHub Actions cron jobs and shipped an installable PWA front end for daily reading – a full round trip from data ingestion to a deployed, user-facing product, built and maintained solo.

Computational Physics Lab — C

- Implemented the Rössler attractor, random walks, percolation, and a lattice-gas model in plain C for a university lab course, with a `make check` target that recomputes reference results and fails the build on numerical drift – an early habit of treating scientific code with the same rigor as production code.